



INDIAN INSTITUTE OF SCIENCE EDUCATION
AND RESEARCH THIRUVANANTHAPURAM

(An Autonomous Institution under MOE, Govt. of India)

Maruthamala. P.O, Vithura, Thiruvananthapuram – 695551

School of Data Science

Computer Organization & Operating Systems

End-Semester Examination, Vasanth 2026

Maximum Marks: 80

Course Code: DSC315

Weightage: 50

Date: 2026-May-02

Instructions:

- Answer all questions. Provide concise and precise responses. Each mark corresponds to approximately 2 minutes of writing.
- Either skip Q4 or skip any two from the remaining questions.

1. (a) What are the advantages and limitations of contiguous memory allocation for processes?
(b) Compare internal and external fragmentation.
(c) When a page fault occurs and no free frame is available, how does the system select a victim frame?
(d) Consider a process with 7 pages labeled from 0 to 6. The page reference string is given as:

1, 3, 0, 3, 5, 6, 3, 1, 2, 3, 6

Assuming that the number of available page frames is 3, compute the number of page faults using the FIFO and LRU (Least Recently Used) page replacement algorithms.

[2+2+2+4=10]

2. Answer the following questions briefly

- (a) Apply Amdahl's Law to determine the speedup when a program is 80% parallel and 20% serial, using 4 cores.
- (b) Describe the issues associated with asynchronous thread cancellation and illustrate with an example.
- (c) Describe where Direct Memory Access (DMA) is useful, and where is it not suitable?

[4+3+3=10]

3. (a) What are zombie and orphan processes? What is the impact on the system when a process becomes a zombie or an orphan?
(b) Describe the two inter-process communication (IPC) models. Consider the producer-consumer problem. Write pseudocode to solve it using both IPC models.

[3+(1+6)=10]

4. Consider a system with three types of resources: Bread, Butter, and Jam. There are three worker processes:

- Worker 1 has an unlimited supply of Bread.
- Worker 2 has an unlimited supply of Butter.
- Worker 3 has an unlimited supply of Jam.

There is one coordinator process. In each round, the coordinator randomly places two of the three resources on a shared table, one by one. The worker who possesses the third missing resource will pick up the two items from the table, prepare a sandwich, and signal the coordinator to proceed to the next round.

Now answer the followings.

- (a) Model this problem as a synchronization problem by identifying processes/threads, shared resources, constraints, synchronization primitives, and interaction rules.
- (b) Write pseudocode using semaphores to coordinate the workers and the coordinator.

Ensure that:

- No deadlock occurs.
 - No worker starves.
 - Only the correct worker proceeds in each round.
- (c) Briefly explain how your solution guarantees correctness.

[5+10+5=20]

5. (a) What is a deadlock? State the four necessary conditions for deadlock.
- (b) What is starvation? How is it different from deadlock? Can deadlock handling techniques lead to starvation? Explain with reasoning.
- (c) Consider a system with 3 resource types A, B, & C having total instances (8, 6, 4). The following snapshot is given:

Process	Allocation			Max		
	A	B	C	A	B	C
P0	1	0	1	3	2	2
P1	2	1	0	4	2	1
P2	1	2	1	5	3	2
P3	0	1	1	2	2	2

Available = (4, 2, 1)

Using Banker's Algorithm, determine whether the system is in a safe state. If yes, find a safe sequence, if no, justify. Show all steps.

[3+3+4=10]

6. (a) Discuss the impact of time quantum on the performance of Round Robin scheduling.
 (b) Compare Round Robin scheduling with Priority Scheduling. Explain under what conditions each performs better.
 (c) Justify, if starvation is possible in any? If so, how can it be prevented?

[2+(2+2)+4=10]

7. Consider the following set of processes:

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	2
P4	3	6
P5	4	3

- (a) Draw the Gantt chart for the following scheduling algorithms:
- First-Come, First-Served
 - Shortest Job First
 - Shortest Remaining Time First
- (b) Compare the performance of the above scheduling algorithms based on the following metrics:
- Average waiting time
 - Average latency

[(2+2+2)+(2+2)=10]

8. (a) State the three requirements for a solution to the critical-section problem.
 (b) Give a solution to the general critical-section problem using `compare_and_swap` that satisfies the three requirements.
 (c) Can deadlock or starvation occur when using `wait()` and `signal()` implemented using semaphore? Provide justification.

[2+5+3=10]

9. Briefly answer the following questions.

- (a) Explain the role of system calls in an operating system.
 (b) State the similarities and differences between processes and programs.
 (c) When an I/O operation is requested, should control return to the user program before or after I/O completion? Justify your answer with reasons.
 (d) How many threads should be in a thread pool? Justify your answer.

[2+2+3+3=10]

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